

Fahim Forhad *Software Engineer*

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🧩 PROFESSIONAL SUMMARY

Full-Stack Software Engineer with 4+ years of experience designing and shipping high performance web applications. Led frontend development for a property platform serving 100,000+ users, reducing page load times by 50% through advanced React optimisation. Brings hands on expertise in React, TypeScript, Node.js, and PostgreSQL alongside a Master of Data Science. Passionate about clean architecture, DevOps automation, and building observable, production ready systems.

📁 PROFESSIONAL EXPERIENCE

Queensland Cyber Infrastructure Federation(QCIF) , Sep 2025 – Dec 2025 | Brisbane, AU
Software Engineer (ML Security Engineer)

- **Federated System Architecture:** Designed and deployed a secure, containerised federated learning environment using Docker and Docker Compose, validating **up to 4 concurrent federated nodes** on a local testbed with a computer vision dataset.
- **Scalable Design Approach:** Architected the system for **horizontal scaling beyond local hardware constraints**, separating server (superlink) and client (supernode) components to support multi-organisation deployments.
- **Environment Automation:** Automated end-to-end provisioning with shell scripts, reducing environment setup time from **several hours to under 30 minutes**.
- **Secure Communication Layer:** Implemented privacy-preserving service communication using **mTLS, OpenVPN, and Keycloak (OIDC)**, enabling ML training without direct data transfer.
- **Performance & Security Evaluation:** Conducted system testing to identify bottlenecks and security risks, producing actionable recommendations for production readiness.
- **Technical Documentation:** Authored comprehensive deployment and security documentation used by the Applied AI team for rollout planning.

Sarina Russo Group , *Software Engineer (Data and Analytics)* May 2025 – Aug 2025 | Brisbane, AU

- **End-to-End ML Pipeline:** Built a complete predictive analytics pipeline (research, feature engineering, training, deployment) to identify at-risk students and support early intervention.
- **Actionable Dashboards:** Designed and delivered an interactive Power BI dashboard used by **academic and support staff**, replacing spreadsheet-based reporting and improving decision turnaround time.
- **Privacy-Safe Validation:** Generated synthetic datasets to validate model behaviour under data privacy constraints, enabling robust pre-production testing.
- **Stakeholder Enablement:** Produced user documentation and led a client handover, allowing continued use without ongoing engineering support.

Gain Solutions , *Frontend Team Lead* Sep 2021 – Nov 2023 | Dhaka, Bangladesh

- **Large-Scale Platform Development:** Led frontend development for **Uniteliving.no** , a property platform serving **650+ partner organisations** and **100,000+ users**, comparable to Airbnb or Realestate.com.au .
- **Core Feature Ownership:** Implemented key platform features including **Google Maps-based property search**, real-time chat, and a comprehensive settings module for partners and administrators.
- **Advanced Performance Optimisation:** Reduced modal and table rendering time from **~7 seconds to ~1.3 seconds** by applying advanced React optimisation techniques (**React.memo, useMemo, useCallback, render-path optimisation**).
- **Dashboard Load Improvements:** Cut partner dashboard load time by **~50%**, directly improving usability, user retention, and partner engagement.
- **Technical Leadership & Mentorship:** Led and mentored a team of **4 frontend developers**, enforcing coding standards and ensuring consistent delivery in an Agile environment.
- **Product & Business Collaboration:** Worked closely with product managers and stakeholders on **application lifecycle decisions, performance trade-offs, and frontend best practices**.

Ideaxen , *Junior software developer* Aug 2020 – Jul 2021 | Dhaka, Bangladesh

- **ERP Development:** Engineered core modules for a comprehensive Enterprise Resource Planning (ERP) solution, specifically focusing on complex inventory management systems.
- **Backend Architecture:** Implemented a **Code First** approach using **C# .NET MVC** to design robust database schemas and business logic layers.
- **Data Integrity:** Developed automated tracking for stock movements, ensuring high data accuracy across multi-warehouse environments.

📁 PROJECTS

Career Scout , Python · FastAPI · Playwright · Celery · Redis · PostgreSQL · Ollama · Prometheus · Grafana · Docker

- **Built a full-stack job-intelligence platform** that scrapes LinkedIn, Seek, and Indeed using Playwright-powered scrapers with anti-bot countermeasures and configurable search profiles.
- **Designed a production-grade FastAPI backend** with async SQLAlchemy, Alembic migrations, Pydantic v2 schemas, and a repository/service layered architecture enforcing clean separation of concerns.
- **Implemented a Celery + Redis task queue with cron-based** scheduling, automatic retry logic, and a three-stage async pipeline: scrape → enrich (heuristic skill extraction) → LLM relevance scoring.
- **Shipped Prometheus metrics and four Grafana dashboards** covering scraper throughput, LLM call latency, enrichment rates, and API performance — all auto-provisioned via Docker Compose.
- **Secured secrets management** using Docker Secrets and mTLS

Wellbeing Ecommerce , *React · TypeScript · Node.js · PostgreSQL · Docker · CI/CD · GitHub Actions*

- **Designed and built a full-stack ecommerce platform** (React/TypeScript, Node.js/Express, PostgreSQL/Prisma) with guest checkout, product search and filtering, shopping cart, order tracking, and server side pagination; live in production at wellbeing.mindinroot.com .
- **Engineered a role-based admin panel** with full CRUD for products, categories, and goals, plus stock level management, order fulfilment tracking, user management, and product image uploads, all protected by dedicated admin authentication middleware.
- **Implemented a production-grade auth system** with JWT access/refresh token rotation, TOTP based Multi-Factor Authentication (via authenticator apps), Bcrypt password hashing, RBAC (user/admin), and rate limited authentication endpoints.
- **Architected the infrastructure with Docker, Nginx reverse proxy, and a CI/CD pipeline** (GitHub Actions); unifying frontend and API under a single origin to eliminate cross origin cookie issues and enabling automated builds through to deployment.
- **Built a guest checkout flow** allowing unauthenticated users to complete purchases with email-only, with order retrieval via email verification which separates guest and authenticated order resolution on the backend.

 SKILLS

Programming languages: JavaScript (ES6+), TypeScript, Python, Bash, C#

Frontend Development: React.js, TypeScript, HTML5, CSS3, SCSS, Tailwind, Responsive Design, Performance Optimization, Unit Testing (Jest, React Testing Library), Component Testing

Backend & Databases: FastAPI, Node, Express, REST API, GraphQL, Authentication & RBAC, PostgreSQL, MongoDB, Redis

DevOps & Engineering Practices: Docker, Docker Compose, Git/GitHub, CI/CD (GitHub Actions), Linux (Ubuntu/Debian), Bash Scripting, Code Reviews, Agile/Scrum, Technical documentations

Data Science & ML: Python (NumPy, Pandas, Scikit-learn, PyTorch), Azure Machine Learning, Power BI, Predictive Modeling, Model Deployment

 EDUCATION

Master of Data Science(Professional), *James Cook University Brisbane*  Mar 2024 – Dec 2025 | Brisbane, AU

- **Completed two industry placements:** Federated Learning (QCIF) and Predictive Analytics (Sarina Russo Group)
- **Relevant Coursework:** Data Mining and Machine Learning, Statistical Modeling, Database Modelling
- **CGPA:** 6.14/7.00

Computer Science and Engineering, Mar 2020 | Dhaka, Bangladesh
American Internation University Bangladesh(AIUB) 

 CERTIFICATES

Node.js microservice certificate - Tecognize Training 